Sandipan Seal

AI / Machine Learning Engineer — LLM & Agentic Systems

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Summary

AI Engineer at aiio GmbH with 4 years building **production LLM and agentic systems** (RAG, multi-agent, knowledge graphs), delivered directly to clients as a **forward-deployed engineer**. Master's research on explainable, LLM-based RF-jamming analysis.

Skills

Languages: Python, TypeScript, JavaScript, Go, C++, SQL
LLM / GenAI: RAG, multi-agent & agentic systems, prompt engineering, tool use / function calling, human-in-the-loop, fine-tuning, **knowledge graphs**, embeddings & vector search, **evaluation (LLM-as-judge, RAGAS)**, **guardrails (PII redaction, prompt-injection defense)**, DSPy, MCP, Claude Agent SDK, AI risk & data privacy (GDPR), **LLMOps**
LLM Frameworks & Providers: LangChain, CrewAI, Hugging Face, sentence-transformers, Jina embeddings, **OpenAI**, Claude, **Ollama / vLLM**
ML / Data Science: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, explainable AI, NLP, computer vision, clustering (PCA/UMAP)
Backend & Web: FastAPI, Pydantic, SQLAlchemy, **Node.js, React.js, Next.js**, Tailwind CSS, **REST APIs**, microservices
Databases: PostgreSQL, Redis, **Qdrant (vector DB)**, **FalkorDB (graph DB, Cypher)**, Neo4j, **MongoDB**, Supabase
Cloud, DevOps & Observability: AWS (Lambda, API Gateway, CloudFront, S3, SQS, Glue, Athena), Terraform, **Docker, Kubernetes, CI/CD (GitHub Actions)**, Render, Vercel, Prometheus, Grafana, **OpenTelemetry, Langfuse**
Workflow & Testing: n8n, Playwright, pytest, Locust, API testing

Experience

AI Engineer — aiio GmbH, Magdeburg, Germany *Oct 2026 – Present*

- Gave clients **70–80% more clean, usable data** by building and deploying end-to-end **RAG-enabled multi-agent systems** that ingest process data, extract workflows, and use LLMs to build **knowledge graphs** and chatbots.
- Cut clients' consulting spend by researching and shipping software products that act as an **automated consultant**, so they no longer hire multiple consultants.
- Turned ambiguous client problems into working solutions as a **forward-deployed engineer**: gathered requirements directly with clients, then designed, built, and deployed the full system.

Working Student, AI/ML Software Development — aiio GmbH *Jan 2025 – Sep 2026*

- Added **AI suggestions, recommendations, and AI-generated descriptions** to aiio's main process-data product and Polaris by building LLM-based features into the live products.
- Accelerated delivery of customer-facing AI tools by building responsive **React.js / Next.js** front-ends integrated with production models.

Senior Software Engineer — Persistent Systems Ltd, Kolkata, India *Sep 2021 – Mar 2024*

- Raised healthcare-diagnostic prediction accuracy by ∼20% for client Adaptive Biotechnologies by designing and deploying **predictive ML models** into production.
- Delivered enterprise features end to end for PerkinElmer by owning **React, .NET/C#, and MongoDB** work from API design to production deployment.
- Cut regression-test cycle time by ∼30% by building automated **QA suites** (Selenium, API testing); twice awarded the **High Five Team Award**.

Selected Projects

Master's Thesis, OVGU Magdeburg
From RF Embeddings to Semantic Reasoning: Explainable Jamming Analysis via LLM-Based Evidence Interpretation

- Raised LLM jamming-regime accuracy from chance (19%) to **56%** (**73%** with a zero-cost deterministic correction rule; best of five LLMs 78%) on a class-balanced 2,500-window test set by replacing salience-ranked embedding evidence with a supervised **four-axis LDA representation**.
- Showed that **100%-grounded citations were not always causally faithful** by designing paired perturbation tests (masking, necessity, sufficiency, counterfactuals, matched random controls) with McNemar/Holm and group-aware bootstrap statistics across five LLMs.
- Established the LLM's real contribution by benchmarking it against conventional ML on identical windows: the same four LDA features without an LLM reach 80–81% (vs. 74% for 768-feature logistic regression), so the LLM adds per-decision explanation, not accuracy.

InferOps-AI — Production LLM Gateway GitHub

- Cut LLM inference cost and blocked unsafe requests by building a gateway with **cost-aware model routing,**

RAG (Qdrant/pgvector), **PII redaction**, and **prompt-injection guardrails**, verified by a deterministic + LLM-as-judge + **RAGAS** eval suite at **>80% test coverage**.
- Shipped to production on **AWS serverless** (Lambda, API Gateway, CloudFront, S3) via **Terraform**, with **Langfuse** tracing and Prometheus/Grafana observability.

AegisOps-AI — Agentic Incident Commander GitHub

- Reduced incident-triage effort by building a **multi-agent** system for automated **root-cause analysis** over logs, metrics, and **Kubernetes** state, with confidence scoring and approval-gated runbooks (**human-in-the-loop**).
- Made incident impact visible by adding SLA tracking, a lifecycle state machine, and service-dependency / blast-radius analysis (FastAPI, Next.js, OpenTelemetry).

DeliveryLens-AI — Delivery Intelligence Platform GitHub

- Surfaced engineering workflow bottlenecks and measured intervention ROI by connecting **Jira and GitLab** to an AWS serverless data lake (Glue, Athena, Step Functions) with **guarded Text2SQL** analytics (DSPy) and human-in-the-loop safeguards.

AI Automation Testing Tool — Multi-Agent QA (CrewAI) GitHub

- Cut manual test creation by building a **CrewAI multi-agent** system (explorer, scenario writer, script generator, auto-repairing runner) that turns natural-language requirements into executable **Playwright** tests, with live SSE progress streaming.

Additional projects: AI Compliance Gatekeeper (**n8n** + FastAPI — PII/policy classification & **data-leakage prevention**) · Mine@TENJI (German parliamentary speech recommender — **SentenceTransformers**, **FAISS**). All on GitHub.

Education

M.Sc. Digital Engineering (Computer Science), OVGU Magdeburg *Apr 2024 – Oct 2026 (expected)*
B.Tech. Electronics & Communication Eng., Narula Institute of Technology *Aug 2017 – Jun 2021*

Awards & Patent

Patent: Low-Cost Automated Bag-Valve-Mask Ventilation Machine — Indian Patent IN 202031037602 A (2020)
Awards: Conquer COVID-19 Hackathon Winner (2020) · Best Paper ×2 (I3SET 2019, NCMR 2020) · High Five Team Award ×2 (Persistent)

Certifications

Claude Code in Action — Anthropic (2026) · Fine-tuning LLMs — Hugging Face (2026) · AI Agent Fundamentals — Databricks (2026) · Certified Ethical Hacker (CEH) — EC-Council (2020)

Languages

Bengali (Native, C2) · English (IELTS, C1) · German (Conversational)